

EXPERIMENT - 6

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6.1.1] Incremented Date

A) ALGORITHM

1. Start.
2. Read day.
3. Read month.
4. Read year.
5. If $\text{year} \leq 0$, print "Invalid Date" and Stop.
6. If $\text{month} < 1$ or $\text{month} > 12$, print "Invalid Date" and Stop.
7. If month is 1, 3, 5, 7, 8, 10, 12 then $\text{max_days} = 31$.
8. Else if month is 4, 6, 9, 11 then $\text{max_days} = 30$.
9. Else if month is 2:

If $(\text{year} \% 400 == 0)$ OR $(\text{year} \% 4 == 0 \text{ AND } \text{year} \% 100 \neq 0)$

$\text{max_days} = 29$

Else

$\text{max_days} = 28$
10. If $\text{day} < 1$ or $\text{day} > \text{max_days}$, print "Invalid Date" and Stop.
11. If $\text{day} < \text{max_days}$ then $\text{day} = \text{day} + 1$.
12. Else

$\text{day} = 1$

If $\text{month} == 12$

$\text{month} = 1$

$\text{year} = \text{year} + 1$

Else

$\text{month} = \text{month} + 1$
13. Print day-month-year in format DD-MM-YYYY.
14. Stop.

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B) PYTHON CODE

```
day = int(input())
month = int(input())
year = int(input())

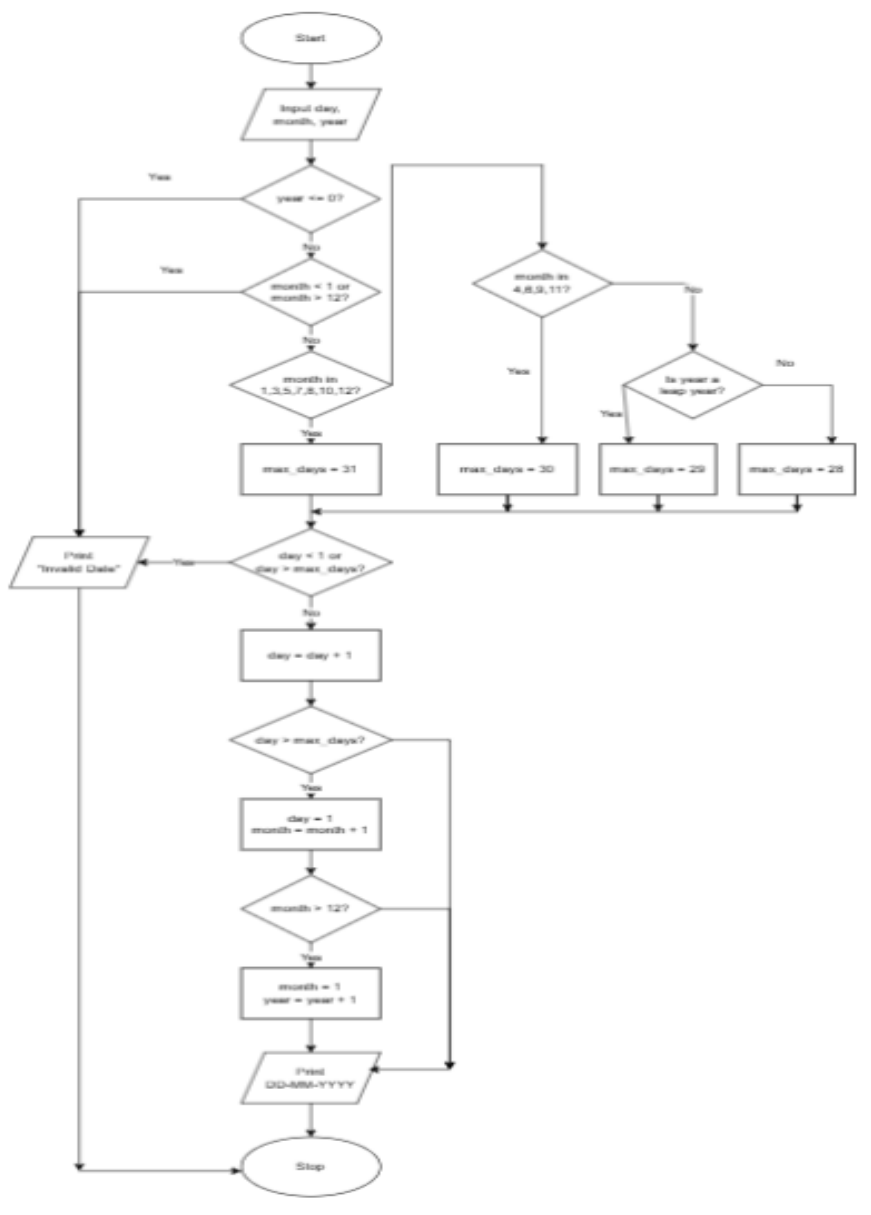
def is_leap(year):
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        return True
    return False

def days_in_month(month, year):
    if month in [1, 3, 5, 7, 8, 10, 12]:
        return 31
    elif month in [4, 6, 9, 11]:
        return 30
    elif month == 2:
        return 29 if is_leap(year) else 28
    else:
        return 0

if year <= 0 or month < 1 or month > 12:
    print("Invalid Date")
elif day < 1 or day > days_in_month(month, year):
    print("Invalid Date")
else:
    day += 1
    if day > days_in_month(month, year):
        day = 1
        month += 1
        if month > 12:
            month = 1
            year += 1
    print(f"{day:02d}-{month:02d}-{year}")
```

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C) FLOWCHART



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D) EXECUTION

CODETANTRA

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6.1.1. Incremented Date

01:29

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

- Valid Month:** 1 to 12
- Valid Day:** Depends on the month and year
 - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
 - April (4), June (6), September (9), November (11): 1 to 30 days
 - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
- Valid Year:** Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format:

<DD>-<MM>-<YYYY>

- If the date is invalid, print: "Invalid Date".

Sample Test Cases

nextDate.py

Submit

Debugger

```
1 # Read input
2 day = int(input())
3 month = int(input())
4 year = int(input())
5
6 # Function to check leap year
7 def is_leap(year):
8     if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
9         return True
10    return False
11
12 # Function to get days in a month
13 def days_in_month(month, year):
14     if month in [1, 3, 5, 7, 8, 10, 12]:
15         return 31
16     elif month in [4, 6, 9, 11]:
17         return 30
18     elif month == 2:
19         return 29 if is_leap(year) else 28
20
21 # Main logic
22 if __name__ == '__main__':
23     day = int(input())
24     month = int(input())
25     year = int(input())
26     days = days_in_month(month, year)
27     if day > days:
28         print("Invalid Date")
29     else:
30         day += 1
31         if month == 12 and day == 31:
32             month = 1
33             year += 1
34             day = 1
35         else:
36             month = month if day < 31 else month + 1
37             day = 1 if day > 31 else day
38         print(f"{day}-{month}-{year}")
```

Average time0.004 sMaximum time0.007 s

4.00 ms7.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 15 ms

Debug

Expected output

Actual output

15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 24 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >